

EXP 3: Implement and demonstrate the working of the Decision Tree based ID3 algorithm. Use an appropriate dataset for building the decision tree and apply this knowledge to classify a new sample.

AIM

To implement and demonstrate the working of the Decision Tree-based ID3 algorithm using an appropriate dataset and classify a new sample.

SOFTWARE USED

- Google Colab
- Python 3.x
- Pandas Library

PROCEDURE

1. Import the required Python libraries.
2. Create or load the dataset.
3. Encode categorical attributes into numerical values.
4. Split the dataset into features and target class.
5. Train the Decision Tree model using the ID3 algorithm (criterion='entropy').
6. Display the generated decision tree rules.
7. Predict the class label of a new sample.
8. Display the prediction result.

GOOGLE COLAB CODE

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, export_text
# Create dataset
data = {
    'Outlook': ['Sunny','Sunny','Overcast','Rain','Rain','Rain',
               'Overcast','Sunny','Sunny','Rain','Sunny',
               'Overcast','Overcast','Rain'],
    'Temperature': ['Hot','Hot','Hot','Mild','Cool','Cool',
                   'Cool','Mild','Cool','Mild','Mild',
                   'Mild','Hot','Mild'],
    'Humidity': ['High','High','High','High','Normal','Normal',
                'Normal','High','Normal','Normal','Normal',
                'High','Normal','High'],
    'Wind': ['Weak','Strong','Weak','Weak','Weak','Strong',
            'Strong','Weak','Weak','Weak','Strong',
            'Strong','Weak','Strong'],
    'PlayTennis': ['No','No','Yes','Yes','Yes','No',
                  'Yes','No','Yes','Yes','Yes',
                  'Yes','Yes','No']
}
# Create DataFrame
df = pd.DataFrame(data)
print("Dataset:\n")
print(df)
# Encode categorical values
encoder = LabelEncoder()
df_encoded = df.copy()
for column in df.columns:
    df_encoded[column] = encoder.fit_transform(df[column])
# Features and Target
X = df_encoded.iloc[:, :-1]
```

```

y = df_encoded.iloc[:, -1]
# Train Decision Tree using ID3
model = DecisionTreeClassifier(criterion='entropy', random_state=0)
model.fit(X, y)
# Display Decision Tree Rules
print("\nDecision Tree Rules:\n")
print(export_text(model, feature_names=list(X.columns)))
# New sample for prediction
new_sample = [[2, 2, 1, 1]]
prediction = model.predict(new_sample)
print("\nPrediction:")\
if prediction[0] == 1:
    print("PlayTennis = Yes")
else:
    print("PlayTennis = No")

```

OUTPUT

Dataset:

	Outlook	Temperature	Humidity	Wind	PlayTennis
0	Sunny	Hot	High	Weak	No
1	Sunny	Hot	High	Strong	No
2	Overcast	Hot	High	Weak	Yes
3	Rain	Mild	High	Weak	Yes
4	Rain	Cool	Normal	Weak	Yes
5	Rain	Cool	Normal	Strong	No
6	Overcast	Cool	Normal	Strong	Yes
7	Sunny	Mild	High	Weak	No
8	Sunny	Cool	Normal	Weak	Yes
9	Rain	Mild	Normal	Weak	Yes
10	Sunny	Mild	Normal	Strong	Yes
11	Overcast	Mild	High	Strong	Yes
12	Overcast	Hot	Normal	Weak	Yes
13	Rain	Mild	High	Strong	No

Decision Tree Rules:

```
|--- Outlook <= 0.50
|   |--- class: 1
|--- Outlook > 0.50
|   |--- Humidity <= 0.50
|       |--- Outlook <= 1.50
|           |--- Wind <= 0.50
|               |--- class: 0
|               |--- Wind > 0.50
|                   |--- class: 1
|           |--- Outlook > 1.50
|               |--- class: 0
|   |--- Humidity > 0.50
|       |--- Wind <= 0.50
|           |--- Temperature <= 1.00
|               |--- class: 0
|               |--- Temperature > 1.00
|                   |--- class: 1
|       |--- Wind > 0.50
|           |--- class: 1
```

Prediction:

PlayTennis = Yes

RESULT

Thus, the Decision Tree based ID3 algorithm was successfully implemented using an appropriate dataset, and the new sample was correctly classified as PlayTennis = Yes.